

HW 8 - Zeshui Song

Notes:

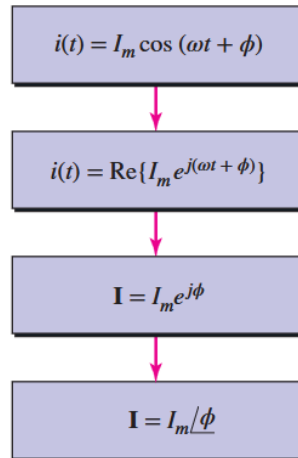
- A sinusoidal forcing function always results in a sinusoidal forced response of the same frequency in a linear circuit. This means that if the input is a sine then the output is a sine of the same frequency, but possibly different amplitude and phase.
- We can rewrite sinusoidal functions as the real and imaginary part of a complex exponential through Euler's identity

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

$$\cos(\omega t + \phi) = \operatorname{Re}\{e^{j(\omega t + \phi)}\}$$

$$\sin(\omega t + \theta) = \operatorname{Im}\{e^{j(\omega t + \theta)}\}$$

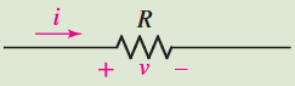
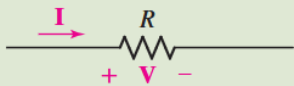
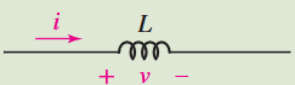
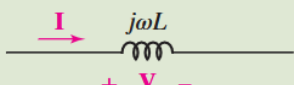
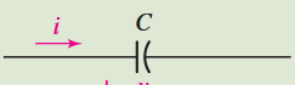
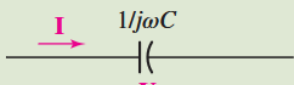
- Since all sinusoidal functions have the same frequency, we can use phasor notation, suppressing the time dependence term $e^{j\omega t}$, resulting in a phasor.



Note that $\mathbf{I}$ represent complex numbers, and we will take the real or imaginary part of the resulting phasor to get to the actual time domain response.

Additionally, differentiation and integration in the time domain correspond to multiplication and division by $j\omega$ in the phasor domain, respectively.

- $\mathbf{V}$ and $\mathbf{I}$ relationships:

Time Domain		Frequency Domain	
	$v = Ri$	$\mathbf{V} = R\mathbf{I}$	
	$v = L \frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$	
	$v = \frac{1}{C} \int i dt$	$\mathbf{V} = \frac{1}{j\omega C} \mathbf{I}$	

We can use these complex valued currents and voltages with KCL and KVL as we normally would to analyze circuits, and then take the real or imaginary part of the resulting phasor to get to the actual time domain response.

- We define **impedance** to be the relationship between the complex voltage and current in a circuit element:

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}}$$

These impedances sum in series and parallel just like resistances and can also be used in nodal and mesh analysis using the relationship $\mathbf{V} = \mathbf{IZ}$ (just like Ohm's law).

- Since $\mathbf{Z}$ is a complex number, we can express it either in rectangular form $\mathbf{Z} = R + jX$ or in polar form $\mathbf{Z} = |\mathbf{Z}|e^{j\theta}$, where R is the real part of the impedance (resistance), and X is the imaginary part of the impedance (reactance).
- Although we have complex voltages and currents for sinusoidal sources, because the circuit is linear, **superposition**, **source transformations**, and **Thevenin/Norton equivalents** still apply in the phasor domain. Just remember to use impedances instead of resistances.

- 10.2** 2. (a) Express each of the following as a single *cosine* function: $300 \sin 628t$, $4 \sin (3\pi t + 30^\circ)$, $14 \sin (50t - 5^\circ) - 10 \cos 50t$. (b) Express each of the following as a single *sine* function: $2 \cos (100t + 45^\circ)$, $3 \cos 4000t$, $5 \cos (2t - 90^\circ) + 10 \sin (2t)$.

We can rewrite this as a single cosine using the trig identities:

$$\sin(\theta) = \cos(\theta - 90^\circ), \quad \cos(\theta) = \sin(\theta + 90^\circ)$$

Note that while it is true that $\sin(\theta) = \cos(90^\circ - \theta)$ and $\cos(\theta) = \sin(90^\circ - \theta)$, the negative sign on θ would change the direction of the phasor from CCW to CW, so to keep consistency, we will try to keep θ positive.

(a) $300 \sin(628t)$

$$300 \sin(628t) = 300 \cos(628t - 90^\circ)$$

(b) $4 \sin(3\pi t + 30^\circ)$

$$4 \sin(3\pi t + 30^\circ) = 4 \cos(3\pi t - 60^\circ)$$

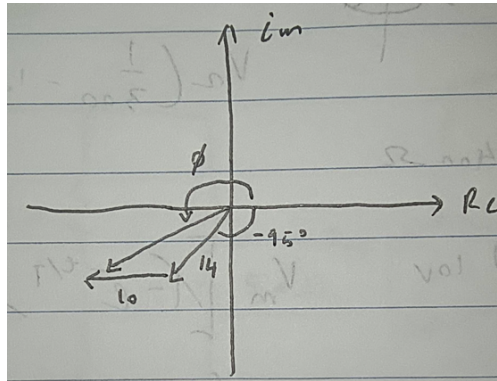
(c) $14 \sin(50t - 5^\circ) - 10 \cos(50t)$

Here it would be useful to perform this addition using phasors. First converting each term to a cosine:

$$14 \sin(50t - 5^\circ) = 14 \cos(50t - 95^\circ)$$

Both have the same frequency, so we can represent them as phasors and add them:

$$14 \cos(50t - 95^\circ) = 14 \angle -95^\circ, \quad -10 \cos(50t) = -10 \angle 0^\circ$$



The sum has magnitude

$$\sqrt{(14 \cos(-95^\circ) - 10)^2 + (14 \sin(-95^\circ))^2} = 17.899$$

and angle

$$\tan^{-1} \left(\frac{14 \sin(-95^\circ)}{(14 \cos(-95^\circ) - 10)} \right) = 51.183^\circ$$

Since $\tan^{-1}$ only gives -90° to 90° , we need to add 180° to get the correct angle

$$\phi = 51.183^\circ + 180^\circ = 231.183^\circ$$

Thus the sum is

$$14 \cos(50t - 95^\circ) - 10 \cos(50t) = \operatorname{Re}(17.899e^{j(50t+231.183^\circ)}) = \boxed{17.899 \cos(50t + 231.183^\circ)}$$

(d) $2 \cos(100t + 45^\circ)$ as a sine:

$$\boxed{2 \cos(100t + 45^\circ) = 2 \sin(100t + 135^\circ)}$$

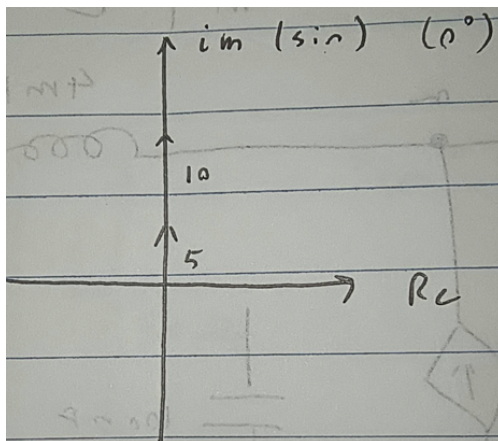
(e) $3 \cos(4000t)$

$$\boxed{3 \cos(4000t) = 3 \sin(4000t + 90^\circ)}$$

(f) $5 \cos(2t - 90^\circ) + 10 \sin(2t)$

First we convert to phasors, using sin as our reference (the positive imaginary axis is now 0°):

$$5 \cos(2t - 90^\circ) = 5 \sin(2t) = 5\angle 0^\circ, \quad 10 \sin(2t) = 10\angle 0^\circ$$



The sum has magnitude

$$\sqrt{(5 + 10)^2} = 15$$

and angle

$$\tan^{-1}\left(\frac{0}{5 + 10}\right) = 0^\circ$$

Thus the sum is

$$5 \sin(2t) + 10 \sin(2t) = \operatorname{Im}(15e^{j(2t+0^\circ)}) = \boxed{15 \sin(2t)}$$

- 10.9** 9. Household electrical voltages are typically quoted as either 110 V, 115 V, or 120 V. However, these values do not represent the peak ac voltage. Rather, they represent what is known as the root mean square of the voltage, defined as

$$\mathbf{V}_{\text{rms}} = \sqrt{\frac{1}{T} \int_0^T \mathbf{V}_m^2 \cos^2(\omega t) dt}$$

where T = the period of the waveform, V_m is the peak voltage, and ω = the waveform frequency ($f = 60$ Hz in North America). (a) Perform the indicated integration, and show that for a sinusoidal voltage,

$$\mathbf{V}_{\text{rms}} = \frac{\mathbf{V}_m}{\sqrt{2}}$$

(b) Compute the peak voltages corresponding to the rms voltages of 110, 115, and 120 V.

Note that while the values are given in the frequency domain (denoted by the bold $\mathbf{V}$), the exponential term $e^{j\phi}$ is constant through time so its just a constant that sticks around.

(a) First evaluating the inside integral:

$$\begin{aligned} \frac{1}{T} \int_0^T V_m^2 \cos^2(\omega t) dt &= \frac{V_m^2}{T} \int_0^T \cos^2(\omega t) dt \\ \int_0^T \cos^2(\omega t) dt &= \int_0^T \frac{\cos(2\omega t) + 1}{2} dt = \left[\frac{t}{2} \right]_0^T + \left[\frac{\sin(2\omega t)}{4\omega} \right]_0^T = \frac{T}{2} + 0 \end{aligned}$$

Note: The $\sin(2\omega t)$ term is 0 because $\omega = \frac{2\pi}{T}$, so $2\omega T = 4\pi$, and $\sin(4\pi) = 0$.

Thus:

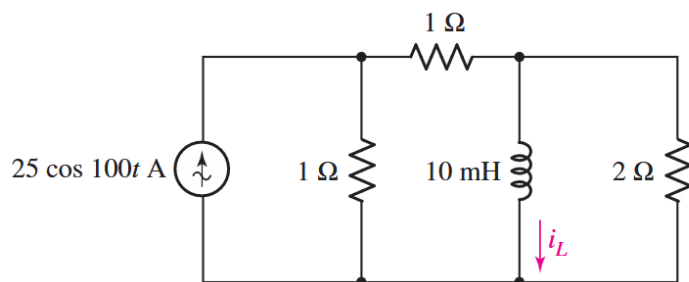
$$\mathbf{V}_{\text{rms}} = \sqrt{\frac{V_m^2}{T} \frac{T}{2}} = \boxed{\frac{V_m}{\sqrt{2}}}$$

(b) To find the peak voltages ($\mathbf{V}_m$), we just need to apply to rms formula:

$$\mathbf{V}_m = \mathbf{V}_{\text{rms}} \cdot \sqrt{2}$$

- 110V rms $\Rightarrow$ 155.56V peak
- 115V rms $\Rightarrow$ 162.63V peak
- 120V rms $\Rightarrow$ 169.71V peak

- 10.11** 11. Assuming there are no longer any transients present, determine the current labeled i_L in the circuit of Fig. 10.47. Express your answer as a single sinusoid.



■ **FIGURE 10.47**

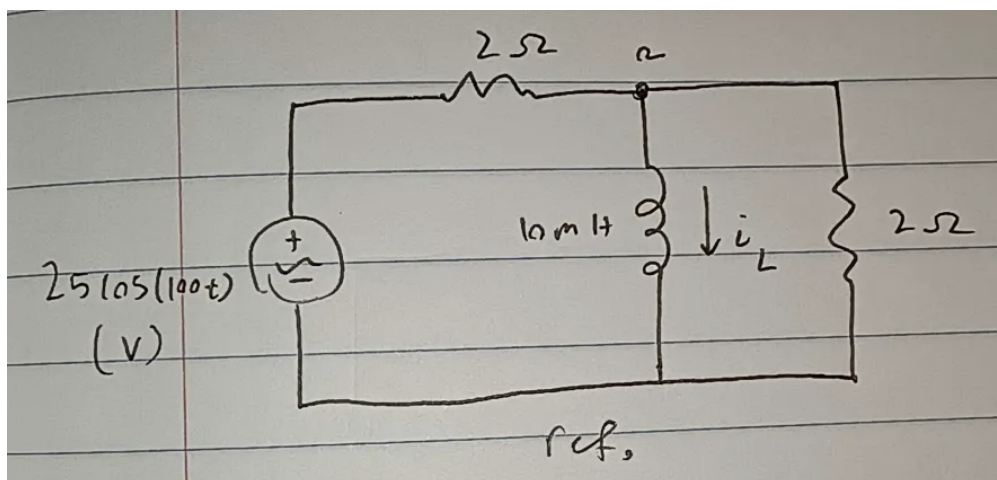
Since this circuit is at sinusoidal steady state, we can use phasor analysis with impedances to solve the problem ($\mathbf{V} = \mathbf{IZ}$).

First, using a source transformation to turn the current source into a voltage source:

$$\mathbf{I}_s = 25\angle 0^\circ, \quad R_p = 1\ \Omega$$

$$\Rightarrow \mathbf{V}_s = \mathbf{I}_s \cdot R_p = 25\angle 0^\circ \cdot [1\ \Omega] = 25\angle 0^\circ$$

Combining the $1\ \Omega$ and $1\ \Omega$ resistors in series:



Using nodal analysis (KCL) at the top node:

$$\frac{\mathbf{V}_s - \mathbf{V}_a}{[2\ \Omega]} = \frac{\mathbf{V}_a}{j[100\ \text{rad/s}] \cdot [10 \times 10^{-3}\ \text{H}]} + \frac{\mathbf{V}_a}{[2\ \Omega]}$$

In rectangular form, $\mathbf{V}_s = 25 + j0$, and $j[100\ \text{rad/s}] \cdot [10 \times 10^{-3}\ \text{H}] = 1j$. Thus:

$$\frac{(25) - \mathbf{V}_a}{2} = \frac{\mathbf{V}_a}{1j} + \frac{\mathbf{V}_a}{2}$$

Solving for $\mathbf{V}_a$:

$$\mathbf{V}_a(1 - j) = \frac{25}{2}$$

Note: $1/j = -j$, we see this by multiplying by j/j .

$$\mathbf{V}_a = \frac{25}{2(1 - j)} = \frac{25}{2} \cdot \frac{1 + j}{(1 - j)(1 + j)} = \frac{25}{2} \cdot \frac{1 + j}{1 + 1} = \frac{25}{4}(1 + j)$$

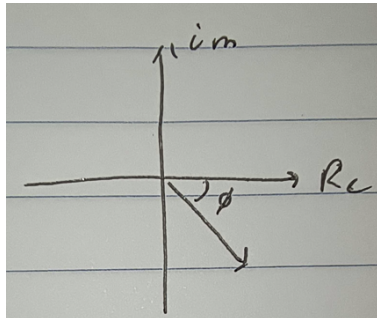
$$\mathbf{V}_a = 6.25 + 6.25j$$

Note: Remember that rectangular form and polar form are equivalent ways of representing phasors. Typically rectangular form is easier for addition and subtraction (sum components), while polar form is easier for multiplication and division (add angles and multiply magnitudes).

Now to find the current through the inductor, we can use $\mathbf{V} = \mathbf{I}\mathbf{Z}$:

$$\mathbf{I}_L = \frac{\mathbf{V}_a}{j1} = \frac{6.25 + j6.25}{j} = -6.25j + 6.25 [\text{A}]$$

Now we can convert from phasor form back to the time domain:

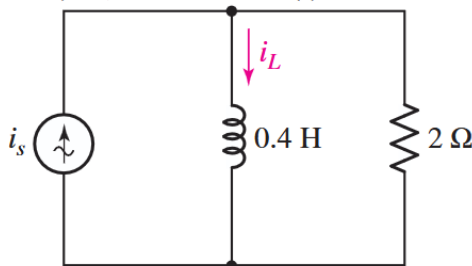


$$|\mathbf{I}_L| = \sqrt{6.25^2 + (-6.25)^2} = 8.838$$

$$\phi = \tan^{-1} \left(\frac{-6.25}{6.25} \right) = -45^\circ$$

$$\mathbf{I}_L = 8.838 \angle -45^\circ = \boxed{8.838 \sin(100t - 45^\circ) [\text{A}]}$$

10.23 23. In the circuit depicted in Fig. 10.51, if i_s is modified such that $i_L(t) = 1.8 \cos(5t + 26.6^\circ)$ A, determine $i_s(t)$.



■ **FIGURE 10.51**

Given the current through the inductor, we can apply current division to solve for $\mathbf{I}_s$:

$$\mathbf{I}_L = 1.8 \angle 26.6^\circ = 1.609 + j0.805$$

Current division says:

$$\begin{aligned} \mathbf{I}_L &= \frac{\frac{1}{\mathbf{Z}_L}}{\frac{1}{\mathbf{Z}_L} + \frac{1}{R}} \mathbf{I}_s = \frac{R}{\mathbf{Z}_L + R} \mathbf{I}_s \\ \Rightarrow \mathbf{I}_s &= \frac{\mathbf{Z}_L + R}{R} \mathbf{I}_L \end{aligned}$$

Note that $\mathbf{Z}_L = j\omega L = j[5 \text{ rad/s}] \cdot [0.4 \text{ H}] = 2j$. Thus:

$$\mathbf{I}_s = \frac{2j + 2}{2} \mathbf{I}_L$$

Since its easier to do multiplication and division in polar form, we can convert the first term to polar form:

$$\frac{2j + 2}{2} = 1 + j$$

Magnitude: $\sqrt{(1)^2 + (1)^2} = 1.4142$

Angle: $\tan^{-1}\left(\frac{1}{1}\right) = 45^\circ$

Thus:

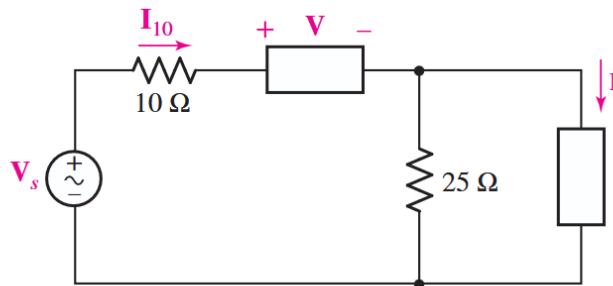
$$\begin{aligned} \mathbf{I}_s &= \frac{\mathbf{Z}_L + R}{R} \mathbf{I}_L = (1.4142 \angle 45^\circ) \cdot (1.8 \angle 26.6^\circ) \\ \mathbf{I}_L &= 2.545 \angle 71.6^\circ \end{aligned}$$

Converting back to time domain:

$\mathbf{I}_L = 2.545 \cos(5t + 71.6^\circ)$

Note: Since the resulting current is a cosine, the forcing function must also be a cosine. So just adding back the time dependence term $e^{j\omega t}$, and taking the real part, we get our solution.

- 10.35** 35. The circuit of Fig. 10.53 is shown represented in the phasor (frequency) domain. If $\mathbf{I}_{10} = 4\angle 35^\circ$ A, $\mathbf{V} = 10\angle 35^\circ$ V, and $\mathbf{I} = 2\angle 35^\circ$ A, (a) across what type of element does $\mathbf{V}$ appear, and what is its value? (b) Determine the value of $\mathbf{V}_s$.



■ **FIGURE 10.53**

- (a) To find the impedance of the element corresponding to $\mathbf{V}$, we can use the relationship $\mathbf{V} = \mathbf{I}\mathbf{Z}$:

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} = \frac{(10\angle 35^\circ)}{(4\angle 35^\circ)} = 2.5\angle 0^\circ$$

Since it has a phase angle of 0° , this means that the element is purely resistive (no imaginary term, so no reactance). Thus the element is a $2.5\ \Omega$ resistor.

- (b) To find $\mathbf{V}_s$, we can use KCL at the top node to find the current entering the $25\ \Omega$ resistor, and then use $\mathbf{V} = \mathbf{I}\mathbf{Z}$ to find the voltage across the $25\ \Omega$ resistor. Then, we can use KVL around the left loop to find $\mathbf{V}_s$:

$$\mathbf{I}_{10} = \mathbf{I}_{25} + \mathbf{I}$$

$$\implies \mathbf{I}_{25} = \mathbf{I}_{10} - \mathbf{I} = (4\angle 35^\circ) - (2\angle 35^\circ) = 2\angle 35^\circ$$

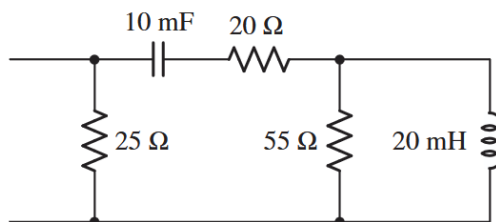
$$\mathbf{V}_{25} = \mathbf{I}_{25} \cdot [25\ \Omega] = (2\angle 35^\circ) \cdot (25\ \Omega) = 50\angle 35^\circ$$

KVL:

$$\mathbf{V}_s - \mathbf{I}_{10} \cdot [10\ \Omega] - \mathbf{V} - \mathbf{I}_{25} \cdot [25\ \Omega] = 0$$

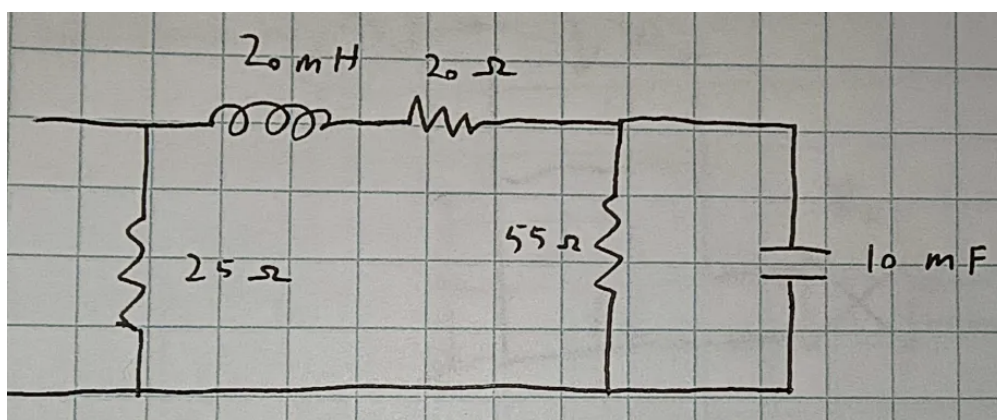
$$\mathbf{V}_s = (4\angle 35^\circ) \cdot (10\ \Omega) + (10\angle 35^\circ) + (50\angle 35^\circ) = \boxed{100\angle 35^\circ}$$

- 10.41** 41. Exchange the capacitor and inductor in the network shown in Fig. 10.54, and calculate the equivalent impedance looking into the open terminals if $\omega = 25 \text{ rad/s}$.



■ **FIGURE 10.54**

First redraw the circuit having exchanged the capacitor and inductor:



Sum the 55Ω resistor in parallel with the 10 mF capacitor:

$$\frac{1}{j\omega C} = \frac{1}{j[25 \text{ rad/s}] \cdot [10 \times 10^{-3} \text{ F}]} = \frac{1}{0.25j} = -4j$$

$$\mathbf{Z} = \frac{(-4j)[55 \Omega]}{(-4j) + [55 \Omega]} = \frac{-220j}{55 - 4j} = \frac{-220j(55 + 4j)}{(55 - 4j)(55 + 4j)}$$

$$\mathbf{Z} = \frac{-12100j + 880}{3041} = 0.289 - 3.978j$$

Sum the 20 mH inductor in series with the 20Ω resistor and $\mathbf{Z}$:

$$j\omega L = j[25 \text{ rad/s}] \cdot [20 \times 10^{-3} \text{ H}] = 0.5j$$

$$\mathbf{Z}_1 = (0.5j) + [20 \Omega] + \mathbf{Z} = 0.5j + 20 + 0.289 - 3.978j$$

$$\mathbf{Z}_1 = 20.289 - 3.478j$$

Finally, we can sum $\mathbf{Z}_1$ in parallel with the 25Ω resistor:

$$\mathbf{Z}_{\text{eq}} = \frac{\mathbf{Z}_1 \cdot [25 \Omega]}{\mathbf{Z}_1 + [25 \Omega]} = \frac{(20.289 - 3.478j)(25)}{20.289 - 3.478j + 25}$$

$$\mathbf{Z}_{\text{eq}} = \frac{507.225 - 86.95j}{45.289 - 3.478j}$$

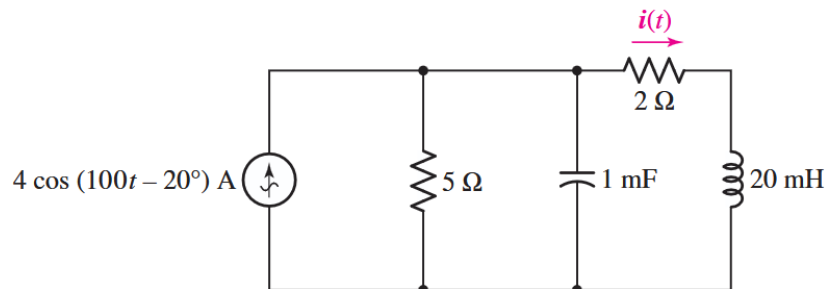
We can divide in polar form more easily, so we can convert the numerator and denominator to polar form:

$$\sqrt{(507.225)^2 + (-86.95)^2} = 514.62, \quad \tan^{-1} \left(\frac{-86.95}{507.225} \right) = -9.727^\circ \quad (\text{numerator})$$

$$\sqrt{(45.289)^2 + (-3.478)^2} = 45.42, \quad \tan^{-1} \left(\frac{-3.478}{45.289} \right) = -4.39^\circ \quad (\text{denominator})$$

$$\mathbf{Z}_{\text{eq}} = \frac{514.62 \angle -9.727^\circ}{45.42 \angle -4.39^\circ} = \boxed{11.33 \angle -5.33^\circ}$$

10.44 44. Employ phasor-based analysis to obtain an expression for $i(t)$ in the circuit of Fig. 10.57.



■ **FIGURE 10.57**

Source:

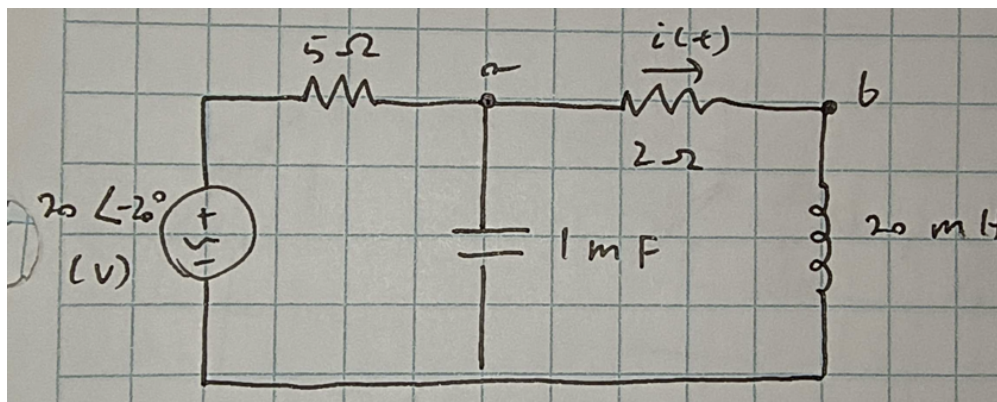
$$\mathbf{I}_s = 4 \angle -20^\circ$$

Source transformation: $R_p = 5 \Omega$:

$$\mathbf{V}_s = \mathbf{I}_s \cdot R_p = (4 \angle -20^\circ) \cdot [5 \Omega] = 20 \angle -20^\circ$$

$$\mathbf{V}_s = 18.793 - 6.84j$$

Redrawing the circuit:



Nodal analysis at node a:

$$\begin{aligned}\frac{1}{j\omega C} &= \frac{1}{j[100 \text{ rad/s}] \cdot [10^{-3} \text{ F}]} = \frac{1}{0.1j} = -10j \\ j\omega L &= j[100 \text{ rad/s}] \cdot [20 \times 10^{-3} \text{ H}] = 2j \\ \frac{(18.793 - 6.84j) - \mathbf{V}_a}{[5 \Omega]} &= \frac{\mathbf{V}_a}{-10j} + \frac{\mathbf{V}_a - \mathbf{V}_b}{[2 \Omega]} \\ \Rightarrow 3.758 - 1.368j &= (0.7 + 0.1j)\mathbf{V}_a - 0.5\mathbf{V}_b\end{aligned}$$

Nodal analysis at node b:

$$\begin{aligned}\frac{\mathbf{V}_a - \mathbf{V}_b}{[2 \Omega]} &= \frac{\mathbf{V}_b}{2j} \\ \Rightarrow 0 &= 0.5\mathbf{V}_a + (-0.5 + 0.5j)\mathbf{V}_b \\ \mathbf{V}_b &= \frac{-0.5}{(-0.5 + 0.5j)}\mathbf{V}_a\end{aligned}$$

Plugging into the first equation and solving for $\mathbf{V}_a$:

$$\begin{aligned}3.758 - 1.368j &= (0.7 + 0.1j)\mathbf{V}_a - 0.5\frac{-0.5}{(-0.5 + 0.5j)}\mathbf{V}_a \\ 3.758 - 1.368j &= (0.7 + 0.1j)\mathbf{V}_a + \frac{0.25(-0.5 - 0.5j)}{(-0.5 + 0.5j)(-0.5 - 0.5j)}\mathbf{V}_a \\ 3.758 - 1.368j &= (0.7 + 0.1j)\mathbf{V}_a + \frac{-0.125 - 0.125j}{0.25 + 0.25}\mathbf{V}_a \\ 3.758 - 1.368j &= (0.7 + 0.1j)\mathbf{V}_a + (-0.25 - 0.25j)\mathbf{V}_a \\ 3.758 - 1.368j &= (0.45 - 0.15j)\mathbf{V}_a \\ \Rightarrow \mathbf{V}_a &= \frac{3.758 - 1.368j}{0.45 - 0.15j}\end{aligned}$$

Using polar form to divide:

$$\sqrt{(3.758)^2 + (-1.368)^2} = 3.99, \quad \tan^{-1}\left(\frac{-1.368}{3.758}\right) = -20.00^\circ \quad (\text{numerator})$$

$$\sqrt{(0.45)^2 + (-0.15)^2} = 0.474, \quad \tan^{-1}\left(\frac{-0.15}{0.45}\right) = -18.43^\circ \quad (\text{denominator})$$

$$\mathbf{V}_a = \frac{3.99\angle -20.00^\circ}{0.474\angle -18.43^\circ} = 8.417\angle -1.57^\circ$$

Plug back into the second equation to find $\mathbf{V}_b$:

$$\begin{aligned}\mathbf{V}_b &= \frac{-0.5}{(-0.5 + 0.5j)}\mathbf{V}_a = \frac{-1}{(-1 + j)}\mathbf{V}_a = \frac{-1(-1 - j)}{(-1 + j)(-1 - j)}\mathbf{V}_a \\ \mathbf{V}_b &= \frac{1 + j}{2}\mathbf{V}_a\end{aligned}$$

Convert to polar form to multiply:

$$0.5 + 0.5j = 0.707\angle 45^\circ$$

$$\mathbf{V}_b = (0.707\angle 45^\circ)(8.417\angle -1.57^\circ) = 5.95\angle 43.43^\circ$$

Finally, we can find the current through the $2\ \Omega$ resistor using $\mathbf{V} = \mathbf{I}\mathbf{Z}$:

$$\mathbf{I} = \frac{\mathbf{V}_a - \mathbf{V}_b}{[2\ \Omega]} = \frac{(8.417\angle -1.57^\circ) - (5.95\angle 43.43^\circ)}{2}$$

Convert to rectangular form to subtract:

$$8.417\angle -1.57^\circ = 8.417 - 0.230j$$

$$5.95\angle 43.43^\circ = 4.32 + 4.09j$$

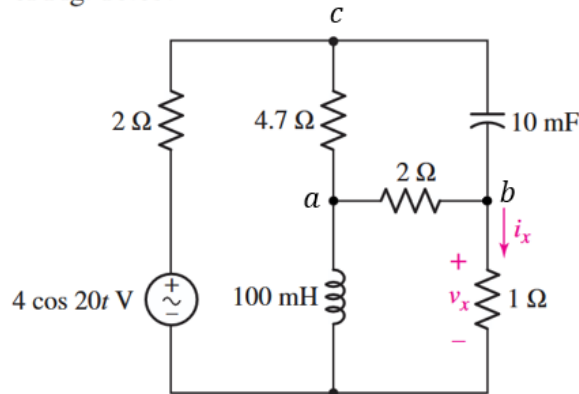
$$\mathbf{I} = \frac{(8.417 - 0.230j) - (4.32 + 4.09j)}{2} = \frac{4.097 - 4.32j}{2} = 2.048 - 2.16j$$

Back to polar form to express the current:

$$\sqrt{(2.048)^2 + (-2.16)^2} = 2.97, \quad \tan^{-1}\left(\frac{-2.16}{2.048}\right) = -46.52^\circ$$

$$\mathbf{I} = 2.97\angle -46.52^\circ \implies \boxed{i(t) = 2.97 \cos(100t - 46.57^\circ)}$$

10.55 55. Employ phasor analysis to obtain an expression for v_x as labeled in the circuit of Fig. 10.63.



■ **FIGURE 10.63** *Ref.*

Start by putting everything in phasor domain:

$$\mathbf{V}_s = 4\angle 0^\circ$$

Impedences:

$$\frac{1}{j\omega C} = \frac{1}{j[20 \text{ rad/s}] \cdot [10 \times 10^{-3} \text{ F}]} = \frac{1}{0.2j} = -5j$$

$$j\omega L = j[20 \text{ rad/s}] \cdot [100 \times 10^{-3} \text{ H}] = 2j$$

Using nodal analysis at node c:

$$\frac{\mathbf{V}_s - \mathbf{V}_c}{[2 \Omega]} = \frac{\mathbf{V}_c - \mathbf{V}_a}{[4.7 \Omega]} + \frac{\mathbf{V}_c - \mathbf{V}_b}{(-5j)}$$

$$\Rightarrow 0.5\mathbf{V}_s = -0.212\mathbf{V}_a + 0.2j\mathbf{V}_b + (0.712 + 0.2j)\mathbf{V}_c$$

Using nodal analysis at node a:

$$\frac{\mathbf{V}_c - \mathbf{V}_a}{[4.7 \Omega]} = \frac{\mathbf{V}_a}{(2j)} + \frac{\mathbf{V}_a - \mathbf{V}_b}{[2 \Omega]}$$

$$\Rightarrow 0 = (0.712 - 0.5j)\mathbf{V}_a - 0.5\mathbf{V}_b - 0.212\mathbf{V}_c$$

Using nodal analysis at node b:

$$\frac{\mathbf{V}_c - \mathbf{V}_b}{(-5j)} + \frac{\mathbf{V}_a - \mathbf{V}_b}{[2 \Omega]} = \frac{\mathbf{V}_b}{[1 \Omega]}$$

$$\Rightarrow 0 = 0.5\mathbf{V}_a + (-1.5 - 0.2j)\mathbf{V}_b + 0.2j\mathbf{V}_c$$

Solving this system of equations gives us:

$$\mathbf{V}_a = 0.7031 + 0.6656j$$

$$\mathbf{V}_b = 0.4093 + 0.5627j$$

$$\mathbf{V}_c = 2.9657 - 0.7498j$$

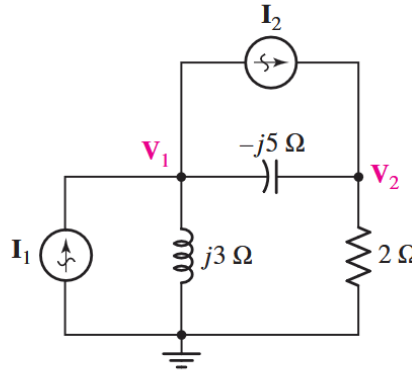
Finally, we can find the current through the inductor using $\mathbf{V}_b$. Putting into polar form:

$$\sqrt{(0.4093)^2 + (0.5627)^2} = 0.695, \quad \tan^{-1}\left(\frac{0.5627}{0.4093}\right) = 53.96^\circ$$

Thus,

$$\mathbf{V}_b = 0.695 \angle 53.96^\circ \implies \boxed{v_x(t) = 0.695 \cos(20t + 53.96^\circ)}$$

10.63 63. The phasor domain circuit of Fig. 10.68 was drawn assuming an operating frequency of 2.5 rad/s. Unfortunately, the manufacturing unit installed the wrong sources, each operating at a different frequency. If $i_1(t) = 4 \cos 40t$ mA and $i_2(t) = 4 \sin 30t$ mA, calculate $v_1(t)$ and $v_2(t)$.



■ FIGURE 10.68

Solving for C and L using the given impedances at $\omega = 2.5$ rad/s:

$$\frac{1}{j\omega C} = \frac{1}{j[2.5 \text{ rad/s}] \cdot C} = -5j \implies C = 0.08 \text{ F}$$

$$j\omega L = j[2.5 \text{ rad/s}] \cdot L = 3j \implies L = 1.2 \text{ H}$$

Impedances at $\omega = 40$ rad/s (i_1):

$$\frac{1}{j\omega C} = \frac{1}{j[40 \text{ rad/s}] \cdot [0.08 \text{ F}]} = \frac{1}{3.2j} = -0.3125j$$

$$j\omega L = j[40 \text{ rad/s}] \cdot [1.2 \text{ H}] = 48j$$

Impedances at $\omega = 30$ rad/s (i_2):

$$\frac{1}{j\omega C} = \frac{1}{j[30 \text{ rad/s}] \cdot [0.08 \text{ F}]} = \frac{1}{2.4j} = -0.4167j$$

$$j\omega L = j[30 \text{ rad/s}] \cdot [1.2 \text{ H}] = 36j$$

(a) Using superposition, we start by isolating i_1 :

$$\mathbf{I}_1 = 4 \angle 0^\circ$$

Using nodal analysis at node 1:

$$\mathbf{I}_1 = \frac{\mathbf{V}_1}{(48j)} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{(-0.3125j)}$$

$$\Rightarrow \mathbf{I}_1 = 3.179j\mathbf{V}_1 - 3.2j\mathbf{V}_2$$

Using nodal analysis at node 2:

$$\frac{\mathbf{V}_1 - \mathbf{V}_2}{(-0.3125j)} = \frac{\mathbf{V}_2}{[2\Omega]}$$

$$\Rightarrow 0 = 3.2j\mathbf{V}_1 + (-0.5 - 3.2j)\mathbf{V}_2$$

Solving this system of equations gives us:

$$\mathbf{V}_1 = 8.091 - 0.916j$$

$$\mathbf{V}_2 = 8.038 + 0.339j$$

Convert to polar and find time domain solution:

$$\sqrt{(8.091)^2 + (-0.916)^2} = 8.142, \quad \tan^{-1}\left(\frac{-0.916}{8.091}\right) = -6.45^\circ$$

$$\sqrt{(8.038)^2 + (0.339)^2} = 8.045, \quad \tan^{-1}\left(\frac{0.339}{8.038}\right) = 2.41^\circ$$

Thus from i_1 :

$$v_{1,1}(t) = 8.142 \cos(40t - 6.45^\circ), \quad v_{2,1}(t) = 8.045 \cos(40t + 2.41^\circ)$$

(b) Now isolating i_2 :

$$\mathbf{I}_2 = 4\angle 0^\circ$$

Using nodal analysis at node 1:

$$\frac{\mathbf{V}_2 - \mathbf{V}_1}{(-0.4167j)} = \mathbf{I}_2 + \frac{\mathbf{V}_1}{(36j)}$$

$$\Rightarrow \mathbf{I}_2 = -2.37j\mathbf{V}_1 + 2.39j\mathbf{V}_2$$

Using nodal analysis at node 2:

$$\mathbf{I}_2 = \frac{\mathbf{V}_2 - \mathbf{V}_1}{(-0.4167j)} + \frac{\mathbf{V}_2}{[2\Omega]}$$

$$\Rightarrow \mathbf{I}_2 = -2.39j\mathbf{V}_1 + (0.5 + 2.39j)\mathbf{V}_2$$

Solving this system of equations gives us:

$$\mathbf{V}_1 = -0.0679 + 1.685j$$

$$\mathbf{V}_2 = -0.0674 - 0.00271j$$

Convert to polar and find time domain solution:

$$\sqrt{(-0.0679)^2 + (1.685)^2} = 1.686, \quad \tan^{-1}\left(\frac{1.685}{-0.0679}\right) = -87.69^\circ$$

Since $\mathbf{V}_1$ is in the second quadrant, we need to add 180° to get the correct angle: $\phi = -87.69^\circ + 180^\circ = 92.31^\circ$.

$$\sqrt{(-0.0674)^2 + (-0.00271)^2} = 0.0675, \quad \tan^{-1}\left(\frac{-0.00271}{-0.0674}\right) = 2.302^\circ$$

Since $\mathbf{V}_2$ is in the third quadrant, we need to add 180° to get the correct angle: $\phi = 2.302^\circ + 180^\circ = 182.3^\circ$. Thus from i_2 :

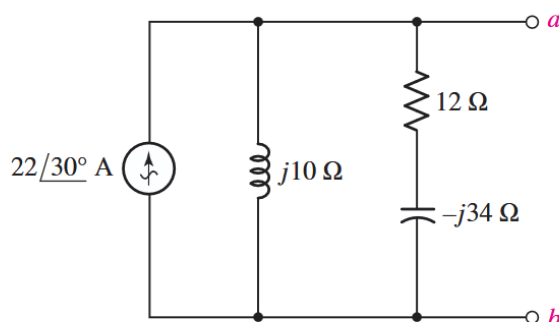
$$v_{1,2}(t) = 1.686 \sin(30t + 92.31^\circ), \quad v_{2,2}(t) = 0.0675 \sin(30t + 182.3^\circ)$$

Finally, we sum the contributions from i_1 and i_2 to get the total voltage at each node:

$$v_1(t) = v_{1,1}(t) + v_{1,2}(t) = 8.142 \cos(40t - 6.45^\circ) + 1.686 \sin(30t + 92.31^\circ) \quad [\text{mV}]$$

$$v_2(t) = v_{2,1}(t) + v_{2,2}(t) = 8.045 \cos(40t + 2.41^\circ) + 0.0675 \sin(30t + 182.3^\circ) \quad [\text{mV}]$$

10.66 66. With regard to the circuit depicted in Fig. 10.70, (a) calculate the Thévenin equivalent seen looking into the terminals marked a and b ; (b) determine the Norton equivalent seen looking into the terminals marked a and b ; (c) compute the current flowing from a to b if a $(7 - j2) \Omega$ impedance is connected across them.



■ FIGURE 10.70

(a) Thevenin equivalent circuit:

Find the equivalent impedance after turning off the source (short circuit for voltage source, open circuit for current source):

Combine $-34j \Omega$ in series with 12Ω :

$$\mathbf{Z}_1 = -34j + 12$$

Combine $\mathbf{Z}_1$ in parallel with $10j \Omega$:

$$\mathbf{Z}_{\text{eq}} = \frac{(12 - 34j)(10j)}{(12 - 34j) + 10j} = \frac{120 - 340j}{12 - 24j}$$

Convert to polar form to divide:

$$\sqrt{(120)^2 + (-340)^2} = 360.55, \quad \tan^{-1} \left(\frac{-340}{120} \right) = -70.56^\circ \quad (\text{numerator})$$

$$\sqrt{(12)^2 + (-24)^2} = 26.83, \quad \tan^{-1} \left(\frac{-24}{12} \right) = -63.43^\circ \quad (\text{denominator})$$

$$\mathbf{Z}_{\text{eq}} = \frac{360.55 \angle -70.56^\circ}{26.83 \angle -63.43^\circ} = 13.44 \angle -7.13^\circ$$

Convert back to rectangular form:

$$\boxed{\mathbf{Z}_{\text{Th}} = \mathbf{Z}_{\text{eq}} = 13.33 - 1.67j}$$

Find the open circuit voltage $\mathbf{V}_{\text{oc}}$:

The voltage across the parallel branches are all equal to

$$\mathbf{V} = \mathbf{I}_s \mathbf{Z}_{\text{eq}} = (22 \angle 30^\circ)(13.44 \angle -7.13^\circ) = 295.68 \angle 22.87^\circ$$

Convert to rectangular form:

$$\boxed{\mathbf{V}_{\text{oc}} = 272.5 + 114.2j}$$

(b) Norton equivalent circuit:

Find the Norton current $\mathbf{I}_N$:

If we short the load (replace with wire), then the current through it is equal to the total source current:

$$\mathbf{I}_{\text{sc}} = \mathbf{I}_s = 22 \angle 30^\circ$$

Convert to rectangular form:

$$\boxed{\mathbf{I}_N = 19.05 + 11j}$$

Find the Norton impedance $\mathbf{Z}_N$:

The Norton impedance is the same as the Thevenin impedance:

$$\boxed{\mathbf{Z}_N = \mathbf{Z}_{\text{Th}} = 13.33 - 1.67j}$$

(c) Find current from a to b if $(7-2j) \Omega$ is connected across it:

Using the Thevenin equivalent circuit, we can use voltage division to find the voltage across the load, and then use $\mathbf{V} = \mathbf{IZ}$ to find the current:

$$\mathbf{V}_{\mathbf{ab}} = \frac{(7 - 2j)}{(7 - 2j) + (13.33 - 1.67j)} \mathbf{V}_{\mathbf{oc}} = \frac{7 - 2j}{20.33 - 3.67j} \mathbf{V}_{\mathbf{oc}}$$

Convert to polar form to divide:

$$\sqrt{(7)^2 + (-2)^2} = 7.28, \quad \tan^{-1} \left(\frac{-2}{7} \right) = -15.95^\circ \quad (\text{numerator})$$

$$\sqrt{(20.33)^2 + (-3.67)^2} = 20.66, \quad \tan^{-1} \left(\frac{-3.67}{20.33} \right) = -10.23^\circ \quad (\text{denominator})$$

$$\mathbf{V}_{\mathbf{ab}} = \frac{7.28 \angle -15.95^\circ}{20.66 \angle -10.23^\circ} \mathbf{V}_{\mathbf{oc}} = (0.352 \angle -5.72^\circ) \cdot (295.68 \angle 22.87^\circ) = 104.1 \angle 17.17^\circ$$

Find current:

$$\mathbf{I}_{\mathbf{ab}} = \frac{\mathbf{V}_{\mathbf{ab}}}{(7 - 2j)} = \frac{104.1 \angle 17.17^\circ}{7.28 \angle -15.95^\circ} = 14.3 \angle 33.12^\circ$$

Thus,

$$\boxed{i_{ab}(t) = 14.3 \cos(100t + 33.12^\circ) [\text{A}]}$$